COLLEGE LIBRARY MANAGEMENT SYSTEM

Java Terminal Application | Project Explanation

Project type	Console-based Java application
Interface	Terminal text menus; no GUI
User roles	Administrator, Librarian, Student
Core rule	Maximum 3 active book borrowings per student

1. Problem Statement

A college library needs an application to manage books and student records, track borrowed and returned books, and limit each student to three active borrowings. If older records exceed this limit, the system should show a warning without deleting those records.

2. Objectives

• Maintain book titles, authors, and categories.

• Register and manage student details.

• Provide role-based access for administrators, librarians, and students.

• Record borrowing and returning activity.

• Search by title/author, filter by category, and sort by title.

• Reject a fourth active borrowing and report existing limit violations.

3. Technology Used

Technology	Purpose
Java	Program logic, classes, menus, and validation
Scanner	Reads terminal input
Map / List collections	Hold user, book, and borrowing records during execution
JDK + Terminal	Compile and run the application

4. User Roles

Role	Permissions
Administrator	Manage books and students; issue/return books; view records and warnings; create, enable/disable, and view accounts.
Librarian	Manage books and student records; issue/return books; view borrowing records and limit warnings.
Student	Browse/search/filter/sort books and view their own borrowing history.

5. Main Features

Book management: Staff can add, edit, browse, search, filter, sort, and deactivate books. A currently borrowed book cannot be deactivated.

Student management: Staff can register students, view details, edit names or student IDs, and disable accounts. Existing borrowing records remain intact.

Borrowing and returns: Issuing a book creates a borrowing record with an ID, student, book, and date. Returning a book marks that record as returned and preserves it as history.

Three-book limit: The program counts active borrowings before issuing a book. If the student already has three, it rejects the request with a clear message. Returned books no longer count.

Existing-record warnings: A report identifies students with more than three active borrowings and warns staff without deleting records.

6. Demo Login Accounts

Role	Username	Password
Administrator	admin	Admin@123
Librarian	librarian	Lib@123
Student	student1	Stu@123
Student	student2	Stu@123

For testing only: Do not use these sample credentials in a real deployment.

7. Sample Books

Title	Author	Category
Clean Code	Robert C. Martin	Programming
Effective Java	Joshua Bloch	Programming
Introduction to Algorithms	Thomas H. Cormen	Algorithms
Artificial Intelligence: A Modern Approach	Stuart Russell	AI
Database System Concepts	Abraham Silberschatz	Database
Computer Networks	Andrew S. Tanenbaum	Networking

8. Compile and Run

Save the file as **LibrarvTerminal.iava** in Downloads, then run these commands in Command Prompt or VS Code terminal:

```
cd /d "%USERPROFILE%\Downloads"
javac LibraryTerminal.java
java LibraryTerminal
```

The JDK must be installed and `java` and `javac` must be available in PATH.

9. Suggested Hackathon Demo

• Log in as librarian and browse the sample books.

• Issue books to student1 until they have three active borrowings.

• Try a fourth issue and show the rejection message.

• Return one book, then issue another to show the limit works.

• Log in as student1 and view personal borrowing history.

• Log in as admin to show account management and the limit-warning report.

10. Important Limitations

• **Temporary storage:** Records are held in memory and reset when the program closes.
• **Plain-text demo passwords:** A real system should hash passwords securely.
• **Single terminal application:** It is not a multi-user network service.
• **Simple dates:** Dates are stored as text for demonstration.

11. Java Concepts Demonstrated

The project uses classes and objects, collections, loops, conditionals, input validation, methods, and role-based menu logic.

12. Conclusion

The College Library Management System provides a terminal-based way to maintain books and student details, record borrowing and returns, enforce the three-book rule, and warn about existing records that exceed the limit. It can later be extended with persistent storage and stronger authentication.